SandgateVP

Linux Installation Handbook

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Revision History

Version	Date	Location	Description	Comments
1.0	09/29/2006		First version	
1.1	10/30/2006			

◆Purpose of this Document

This document is a handbook describing the procedure for running Linux on SandgateVP.

◆Related Documents

None.

◆Points to Remember

- This document may be revised without advance notice. Please check the version No.
- **The Company makes no operation warranty** for any method other than that described in this document.
- Use the method described in this document to check driver operation.
  Technical support for other than this method **will be provided at an additional charge.**

◆Points to Consider

None.

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Chapter 1 Incorporation Procedure Summary

1.1 Items Required to Construct the Environment

The items required for system construction are given below.

Table 1 CD-ROM

File Summary	File Name
Kernel patch for SandgateVP	patch-linux-2.6.17-SGVP-kernel.v001-002 (CD-ROM storage location: /sgvp-patch/kernel)
Redboot patch for SandgateVP	patch_redboot-SGVP.v001-002 (CD-ROM storage location: /sgvp-patch/ redboot)
Kernel binary	zImage (CD-ROM storage location: /binary/ kernel)
Driver binary	modules.tar.bz2 (CD-ROM storage location: /binary/ kernel)
Redboot binary	Redboot.bin (CD-ROM storage location: /binary/ redboot)
Root file system jffs2 image	kompressor.jffs2 (CD-ROM storage location: /binary/ rootfs)
WATCHPOINT related file	FMCLEAR.bat (CD-ROM storage location: /WATCHPOINT)
Driver operation procedure details	(CD-ROM storage location: /check_dirver)

- Things to prepare in advance

Table 2 Development Terminal Types

Name	Comments
PC for cross development	Fedora Core5
PC for image transmission	Windows XP or Windows 2000 (PC on which WATCHPOINT PXA27x can run)
WATCHPOINT PXA27x for Win32 EJ-Debug for PX27x	SandgateVP debugging environment
Console output destination PC	PC that can be used as an RC232C terminal (This can also be used at the PC for cross development or PC for image transmission.)
Serial connection device for SandgateVP	For console output (Required for Redboot, kernel, and root file system installation)
Straight cable	For connecting to the console output destination PC

1.2 System Summary

The configuration for this system is shown in Table 3.

Table 3 System Configuration

	Name and Type
Target board	SandgateVP
Boot loader	Redboot
Kernel	Linux-2.6.17
Root file system	jffs2 image in the flash memory

1.3 System Construction Procedure

In this handbook, the system construction flow is explained in the following order.

1. Preparation of PC for cross compiling
2. Preparation of PC for image transmission
3. Redboot construction method
4. Kernel construction method
5. File system image construction method
6. Linux operation method

Chapter 2 Preparation of PC for Cross Development

This section explains the preparation of the PC for cross development required for system construction.

2.1 Usage Purpose

The PC to be prepared here uses the following environments.

- Redboot build environment
- Kernel build environment
- jffs2 image root file system creation environment

2.2 Environment Requirements for PC for Cross Development

The environmental requirements for the PC used for cross development are shown in Table 7.

Table 4 Environmental Requirements for PC for Cross Development

Item	Requirement
OS	Fedora Core5

2.3 Installing the Operating System in the PC

The customer is asked to perform the operating system installation.
The Fedora Core5 iso image file can be downloaded form the following URL.

Reference URL
Fedora Core5: http://download.fedora.redhat.com/pub/fedora/linux/core/5/i386/iso/

2.4 Program Deployment

This section explains the deployment method for the patch files and binary files required for preparation of the PC for cross development.

* This document presupposes that deployment takes place under /home/sgvp. When installing the files in a directory of another name, appropriately substitute the correct directory name when reading this explanation.

2.4.1 Files to be Deployed

Table 5 List of Files to be Deployed

File Summary	File Name
Kernel patch for SandgateVP	patch_linux2.6.17-SGVP.v001-002
Linux 2.6.17 source	linux-2.6.17.tar.bz2
Redboot patch for SandgateVP	patch_redboot-SGVP.v001-002
Redboot source for SG2P	sg2pxa270-ecos-source.tar.bz2
Root file system	rootfs-xscale-060924-2.tar.bz2
Cross compiler	xscale-060924-2.tar.bz2
Redboot cross compiler	ecoscentric-gnutools-arm-elf-1.4-2.i386linux.tar.bz2

2.4.2 Deployment Procedure

Use the following procedure to deploy the files.

- Kernel source
  In this system construction “linux-2.6.17.tar.bz2” is used. The kernel source can be downloaded from the following URL.

  URL: http://www.kernel.org

  Deploy the kernel source under /home/sgvp/work.
  Execute the following command.

```
$ cd /home/sgvp/work/
$ tar jxvf linux-2.6.17.tar.bz2
```

  Deploy the kernel source under /home/sgvp/work/linux-2.6.17.

- Kernel patch for SandgateVP
  Apply the kernel patch for SandgateVP.
  Execute the following commands.

```
$ cd /home/sgvp/work/linux-2.6.17
$ patch -p1 < patch_linux2.6.17-SGVP.v001-002
```

- Redboot source for SandgateVP
  Deploy the Redboot source for SandgateVP under /home/sgvp/work/redboot.
  Execute the following commands.

```
$ cd /home/sgvp/work/redboot
$ tar jxvf sg2pxa270-ecos-source.tar.bz2
$ cd /home/sgvp/work/redboot/ecos
$ tar zvxf add.tar.gz
```

  Deploy the Redboot source under /home/sgvp/work/redboot.

- Redboot patch for SandgateVP
  Apply the Redboot patch for SandgateVP.
  Execute the following commands.

```
$ cd /home/sgvp/work/redboot/ecos
$ patch -p1 < patch-redboot-SGVP.v001-002
```

- Root file system
  Deploy the root file system under /opt/target.
  If the /opt/target directory does not exist, use the mkdir command to create a directory.
  Use root privileges to execute the following commands.

```
# mkdir /opt/target
# cd /opt/target
# tar jxvf rootfs-xscale-060924-2.tar.bz2
```

  Deploy the root file system under /opt/target/rootfs-xscale-060924-2.

- Cross compiler
  Deploy the cross compiler under /opt/.
  Use root privileges to execute the following commands.

```
# cd /opt
# tar jxvf xscale-060924-2.tar.bz2
```

  Deploy the cross compiler under /opt/ xscale-060924-2/.

- Redboot cross compiler
  Deploy the Redboot cross compiler under /home/sgvp/work/redboot.

```
# cd /home/sgvp/work/redboot
# tar xfj ecoscentric-gnutools-arm-elf-1.4-2.i386linux.tar.bz2
```

  Deploy the Redboot cross compiler under /home/sgvp/work/redboot/.

Chapter 3 Preparation of PC for Image Transmission

This section explains the preparation of the PC required for Redboot, kernel, and root file system writing.

3.1 Usage Purpose

The PC prepared here is used for writing the Redboot, kernel, and root file system to the SandgateVP Flash ROM.

3.2 Environment Requirements for PC for Image Transmission

The PC environmental requirements for image transmission are shown in Table 9.

Table 6 Environmental Requirements for the Host PC

Item	Requirement
OS	Windows XP or Windows 2000
Required port	Has a USB port

3.3 WATCHPOINT PXA27x Program Installation

The Redboot, kernel, and root file system are written to WATCHPOINT and transmitted.

3.3.1 WATCHPOINT Settings File for SandgateVP

The WATCH POINT settings file is stored in the CD-ROM.

Table 7 WATCHPOINT settings file for SandgateVP

File Summary	File Name
Redboot area initialization batch file	FMCLEAR.bat

Chapter 4 Redboot Construction Method

This section explains the Redboot construction procedure.

4.1 Redboot Binary Creation Procedure

The Redboot binary is built on the **PC for cross development** using the ecosconfig command.

Execute the following commands.

```
$ mkdir /home/sgvp/work/redboot/build
$ cd /home/sgvp/work/redboot/build
$ export ECOS_REPOSITORY=/home/sgvp/work/redboot/ecos/packages
$ export PATH=/home/sgvp/work/redboot/gnutools/arm-elf/bin:$PATH
$ ../ecos/configtools/bin/ecosconfig new pxa270sgvp redboot
U CYGHWR_REDBOOT_ARM_LINUX_EXEC_ADDRESS_DEFAULT,
new inferred value 0xA0008000
U CYGSEM_HAL_ROM_MONITOR, new inferred value 1
U CYGDBG_HAL_COMMON_CONTEXT_SAVE_MINIMUM, new inferred
value 0
U CYGDBG_HAL_DEBUG_GDB_INCLUDE_STUBS, new inferred value 1
U CYGPKG_NS_DNS_BUILD, new inferred value 0
U CYGFUN_LIBC_STRING_BSD_FUNCS, new inferred value 0
$ ../ecos/configtools/bin/ecosconfig import ../ecos/redboot-SGVP.ecc
$ ../ecos/configtools/bin/ecosconfig tree
$ make
```

Generate the Redboot image (redboot.bin) under /home/sgvp/work/redboot/build/install/bin.

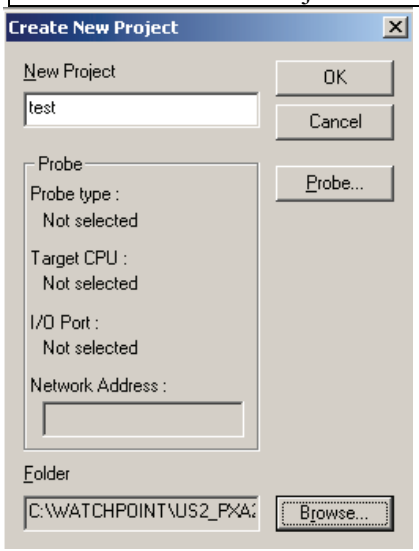
4.2 Method for Writing to Flash ROM

Redboot is written using WATCHPOINT. It is done using the following procedure.

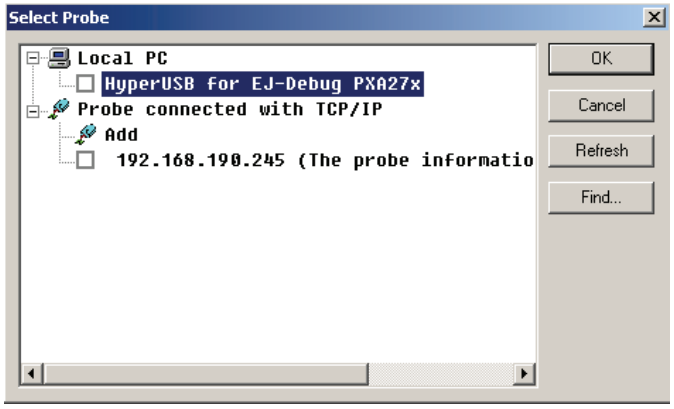
1. Start up WATCHPOINT.

Start up WATCHPOINT PXA27x. Following is and example of test project creation.

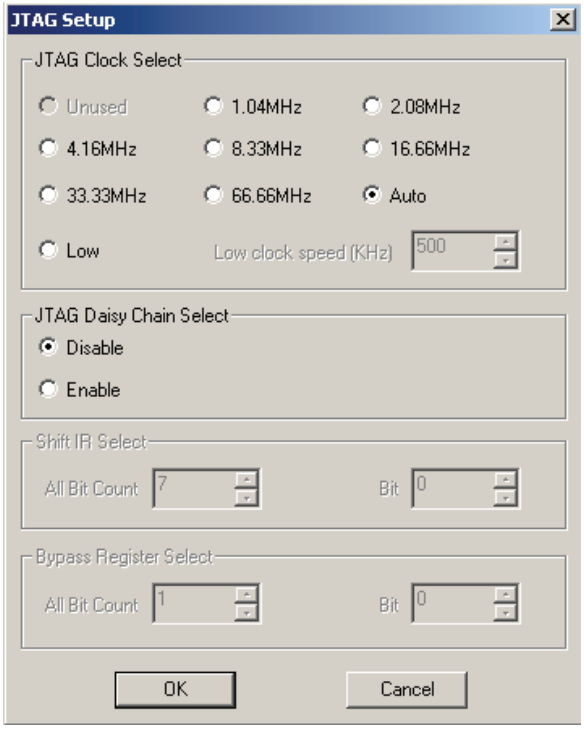
File >> Create New Project

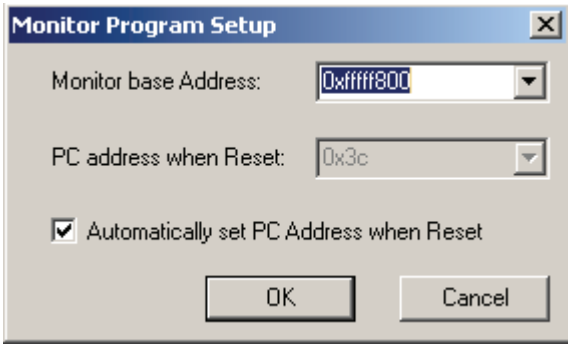


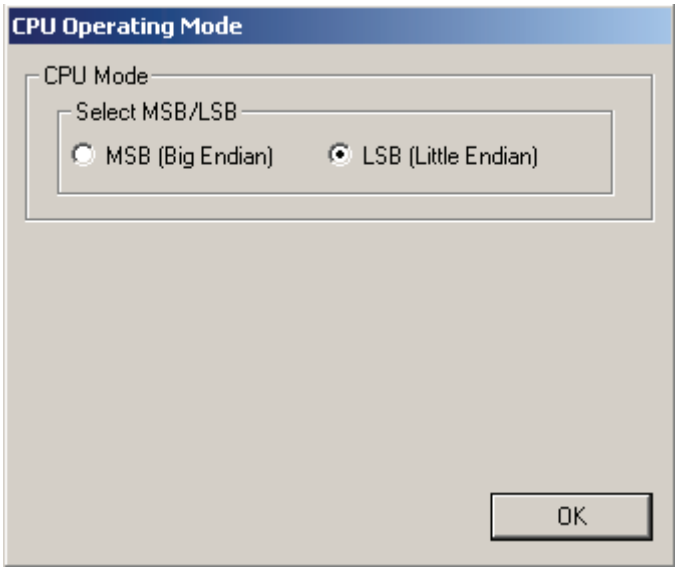
Press the Probe button, select HyperUSB for EJ-Debug PXA27x, and press the OK button.

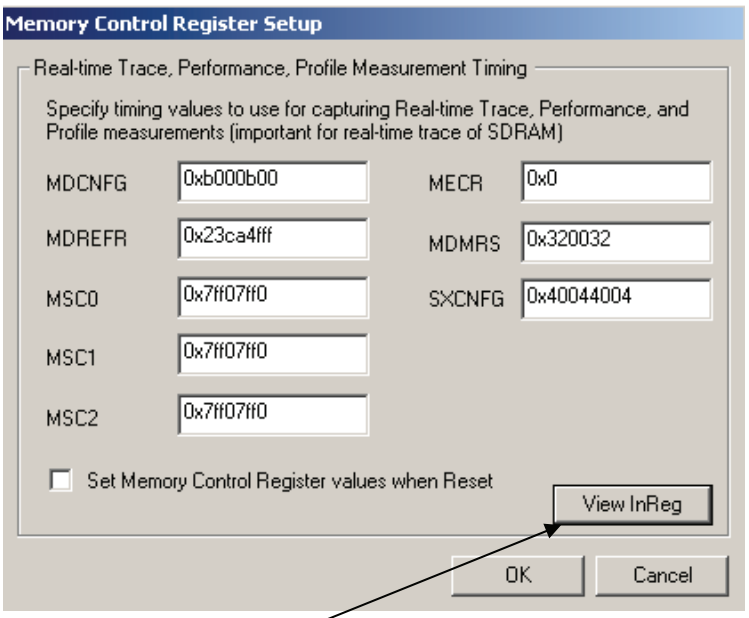


You will be asked regarding the settings. Set them as follows.

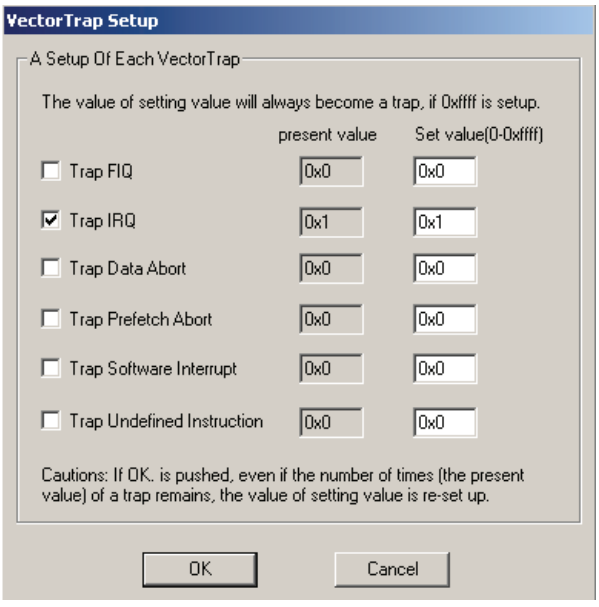








Click the button View InReg to display the value.

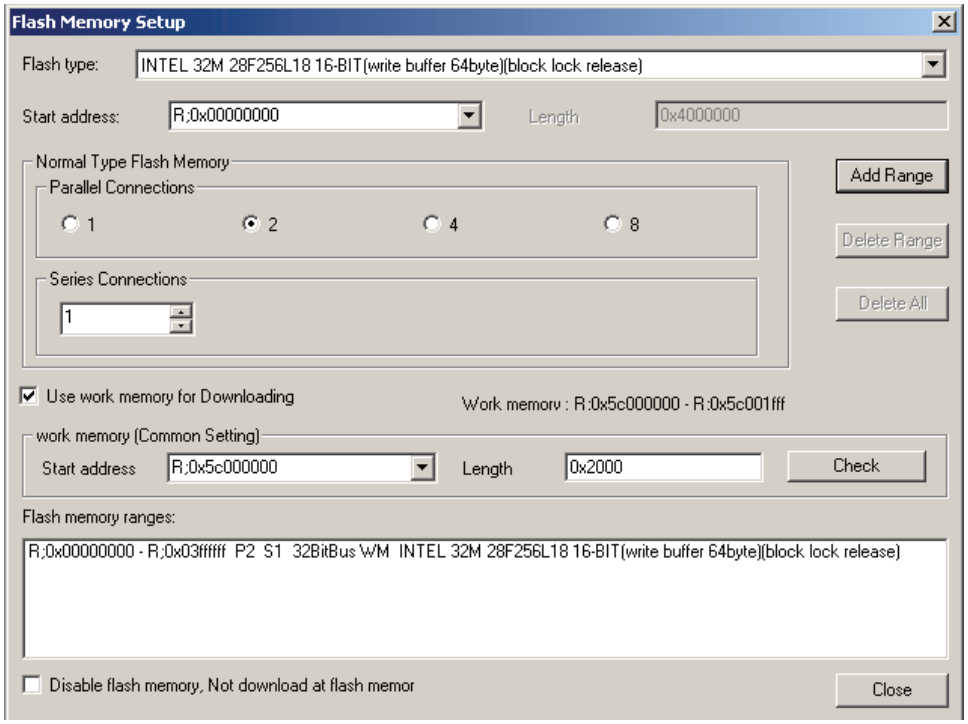


2. Flash settings

Resource >> Flash memory >> Flash settings

Flashtype SGVP	INTEL 32M 28F256L18 16-BIT(write buffer 64byte)(block lock relase)
Start address	R;0x0
Parallel Connections	2
Series Connections	1
Use work memory for	check
Start address	R;0x5c000000
Length	0x2000

After making the settings like those shown in the table above, press the Add button, and then confirm that the settings are added to the settings list.



3. Flash clear

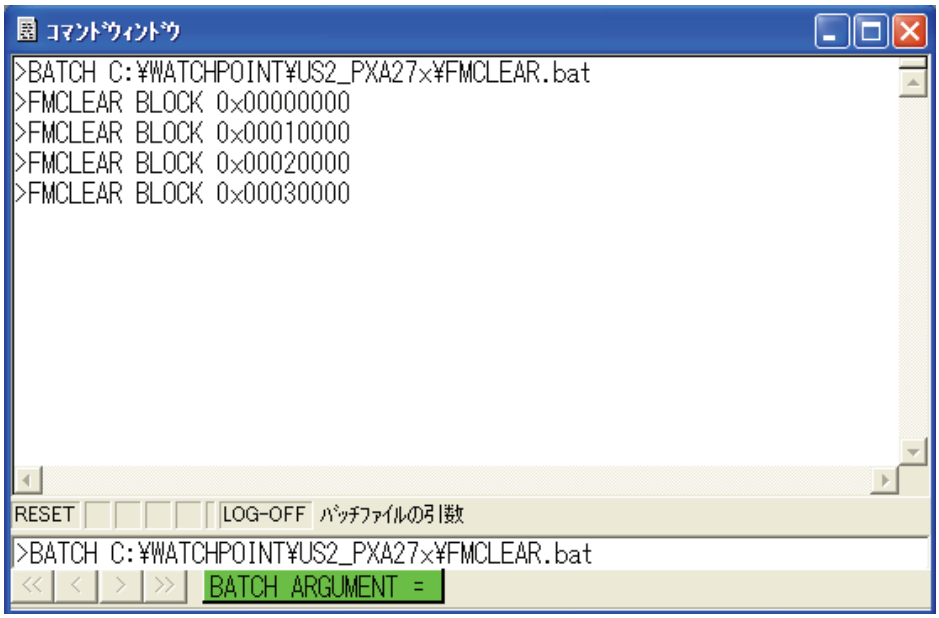
Use FMCLEAR.bat to clear the redboot area.

Copy CD:¥WATCHPOINT¥FMCLEAR.bat to C:¥WATCHPOINT¥US2_PXA27x.

While in WATCHPOINT press the CMD button to open the Command window.



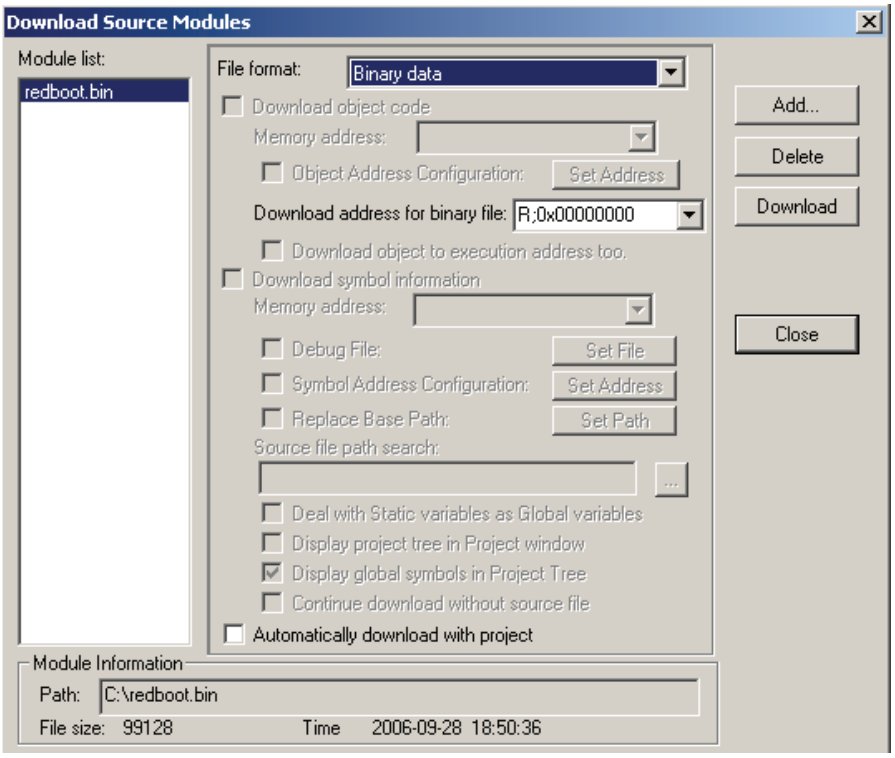
From the Command window, use the BATCH command to execute the batch file.



4. Redboot download

Resource >> Download

Open the Download dialog box, specify a file as shown below, set the format to binary and the address to R;0x00000000, and press the Download button.



4.3 Redboot Startup

This section explains the Redboot startup procedure.

4.3.1 Environment during Redboot Startup

Check the following.

PC for console output

- The serial cable is connected.
- The terminal software is running. (Baud rate: 115,200, data: 8-bit, parity: none, stop bit: 1 bit)

SandgateVP

- The serial cable is connected to the serial connection unit for SandgateVP.

4.3.2 Redboot Startup

If there is no problem with the above environment, turn on the power to SandgateVP.
Redboot will start up and the following message will be output to the terminal.

```
+
RedBoot(tm) bootstrap and debug environment [ROM]
Non-certified release, version Tiger.v004-004 - built 13:06:52, Sep 27 2006

Platform: PXA270 Sandgate-VP (XScale)
Copyright (C) 2000, 2001, 2002, Red Hat, Inc.

RAM: 0xa0000000-0xa4000000, 0xa000e1a8-0xa3f9d000 available
FLASH: 0x10000000 - 0x18000000, 512 blocks of 0x00040000 bytes each.
RedBoot>
```

4.3.3 Initialization of Partition for Redboot

Initialize a partition for Redboot.

Execute the following commands.

```
RedBoot> fis init -f
About to initialize [format] FLASH image system - continue (y/n)? y
*** Initialize FLASH Image System
... Unlock from 0x10040000-0x17f80000:
... Erase from 0x10040000-0x17f80000:
... Unlock from 0x17fc0000-0x17fc0000:
... Erase from 0x17fc0000-0x17fc0000:
... Unlock from 0x18000000-0x18000000:
... Erase from 0x18000000-0x18000000:
... Unlock from 0x17fc0000-0x18000000: .
... Erase from 0x17fc0000-0x18000000: .
... Program from 0xa3f9f000-0xa3fdf000 at 0x17fc0000: .
... Lock from 0x17fc0000-0x18000000: .
RedBoot> fis create "zImage" -n -f 0x13240000 -l 0x00200000 -r 0xa1000000
RedBoot> fis create "jffs2" -n -f 0x10040000 -l 0x03200000
```

The initialized Redboot partition information can be checked using the fis list command.

```
RedBoot> fis list
Name              FLASH addr  Mem addr    Length      Entry point
RedBoot           0x10000000  0x10000000  0x00040000  0x00000000
RedBoot config    0x17F80000  0x17F80000  0x00001000  0x00000000
FIS directory     0x17FC0000  0x17FC0000  0x00040000  0x00000000
zImage            0x13240000  0xA1000000  0x00200000  0x00000000
jffs2             0x10040000  0x10040000  0x03200000  0x00000000
RedBoot>
```

Chapter 5 Kernel Construction Method

This section explains the method for constructing a Linux kernel for SandgateVP.

5.1 Setting the Path to the Cross Compiler

Set the path to the cross compiler.
Set the path as shown below.

```
$ export ARCH=arm
$ export CROSS_COMPILE=/opt/ xscale-060924-2/i686-pc-linux-gnu/bin/ arm-xscale-linux-gnueabi-
```

* The above path setting is only valid within the terminal that made the setting. The path must be set again to perform the same operation using a different terminal.

5.2 Kernel Construction Procedure

This section explains the procedure for creating a kernel image.

5.2.1 Kernel Configuration

Use the following procedure to create the kernel configuration.

1. Move to the kernel source directory.

```
$ cd    /home/sgvp/work/linux-2.6.17
```

2. Copy the configure definition file for SandgateVP.

```
$ cp    arch/arm/configs/sandgatevp-defconfig    .config
```

3. Execute “make menuconfig” to enable the .config file contents.

```
$ make    menuconfig
```

When “menuconfig” is executed, the display changes to the following configuration screen.

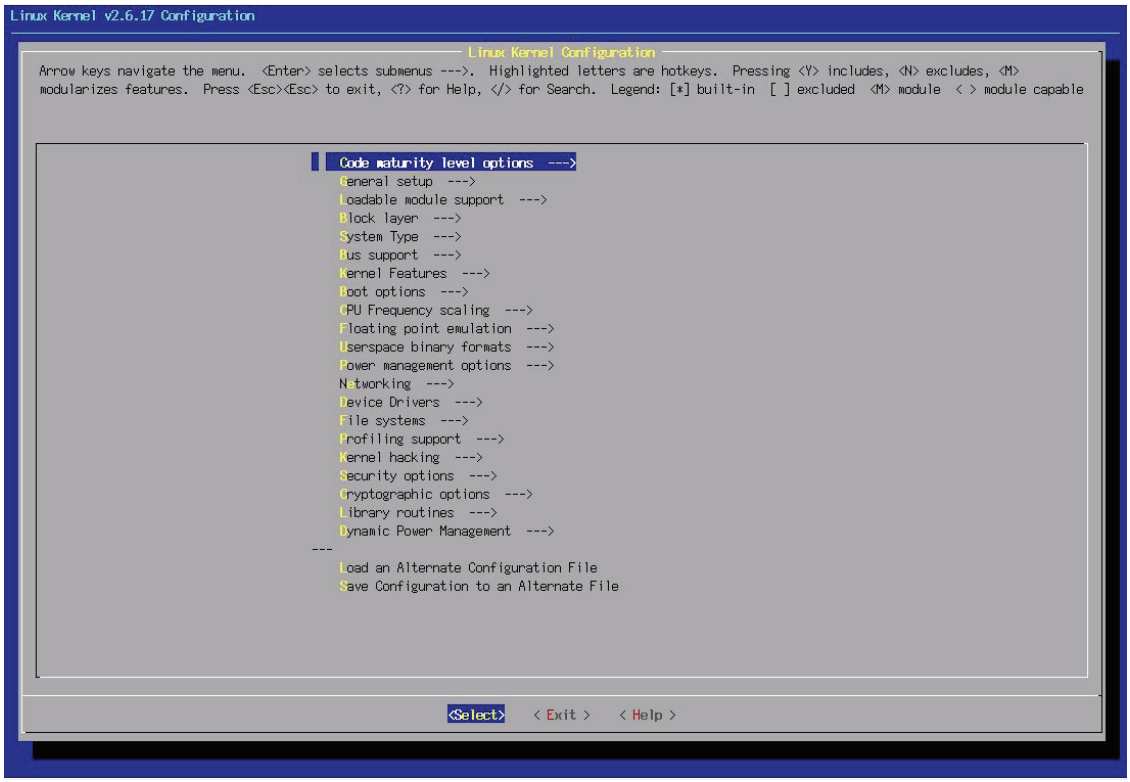


Fig. 1 Kernel configuration screen

Use the arrow keys to move the cursor to <Exit> at the bottom of the screen, and then press the Enter key.

When <Exit> is selected, the display changes to the following screen.

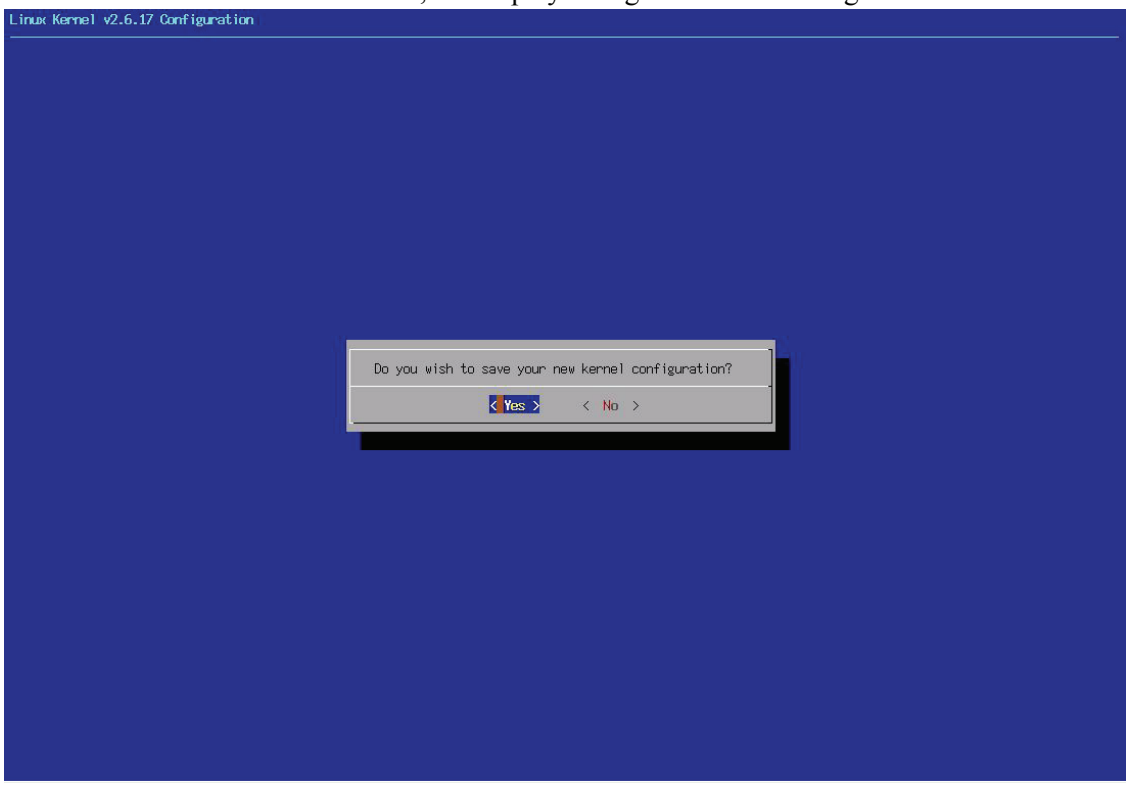


Fig. 2 Kernel configuration exit screen

Use the arrow keys to move the cursor to <Yes> and then press the Enter key.
The Configuration screen will close and the kernel configuration contents will be reflected in the .config file.

5.2.2 Kernel Building

Use the following procedure to build the kernel.

1. Execute the “make” command.

```
$ make
```

When the “make” command is executed, the following message is displayed and kernel building is started.

```
$ make
  CHK     include/linux/version.h
  HOSTCC  scripts/basic/fixdep
  HOSTCC  scripts/basic/split-include
  HOSTCC  scripts/basic/docproc
  HOSTCC  scripts/kconfig/conf.o
  HOSTCC  scripts/kconfig/kxgettext.o
  HOSTCC  scripts/kconfig/mconf.o
  HOSTCC  scripts/kconfig/zconf.tab.o
  HOSTLD  scripts/kconfig/conf
scripts/kconfig/conf -s arch/arm/Kconfig
drivers/i2c/chips/Kconfig:129:warning: 'select' used by config symbol
'SENSORS_ADCM2650' refer to undefined symbol 'I2C_SENSOR'
#
# using defaults found in .config
#
  SPLIT   include/linux/autoconf.h -> include/config/*
  SYMLINK include/asm-arm/arch -> include/asm-arm/arch-pxa
  Generating include/asm-arm/mach-types.h
  CC      arch/arm/kernel/asm-offsets.s
  GEN     include/asm-arm/asm-offsets.h
  CC      scripts/mod/empty.o
  HOSTCC  scripts/mod/mk_elfconfig
  MKELF   scripts/mod/elfconfig.h
  HOSTCC  scripts/mod/file2alias.o
  HOSTCC  scripts/mod/modpost.o
```

The kernel is created below.
Compressed kernel (zImage)
/home/sgvp/work/linux-2.6.17/arch/arm/boot/zImage

2. Install the module in the root file system. Use the root privileges to execute the following command.

```
# make modules_install INSTALL_MOD_PATH=/opt/target/rootfs-xscale-060924-2
```

The driver module is created below.

```
Driver Module
  Under /opt/target/rootfs-xscale-060924-2/lib/modules
```

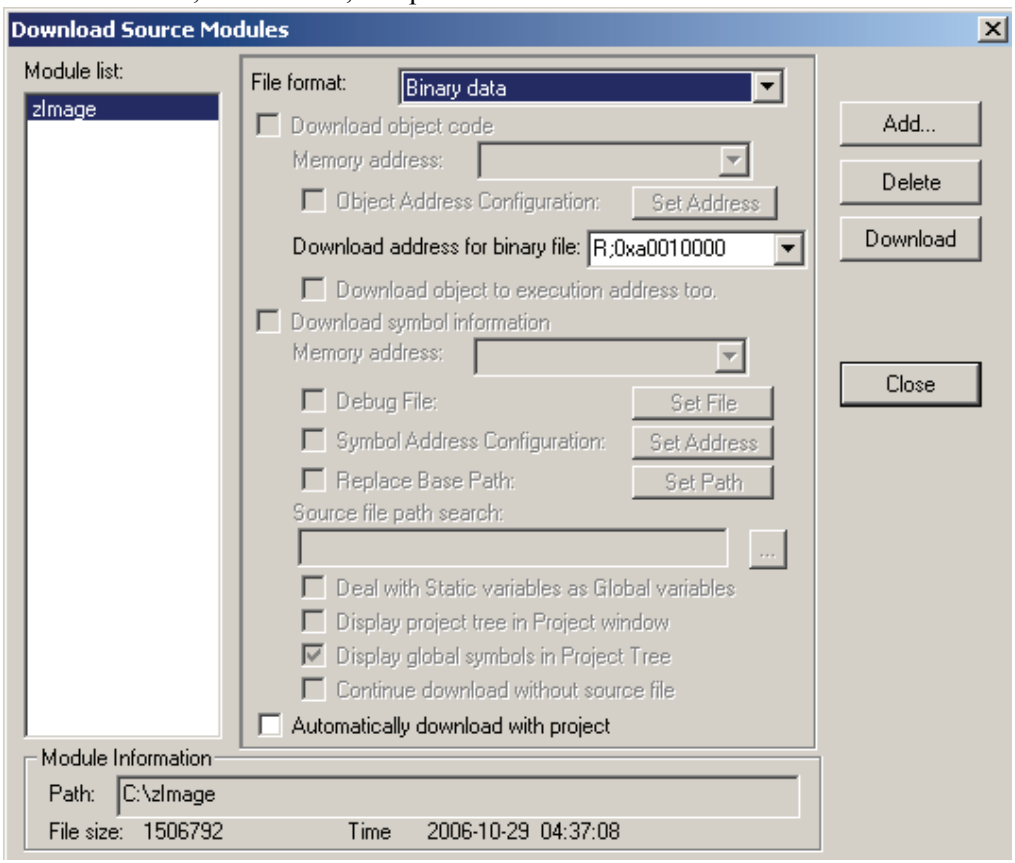
5.3 Method for Writing to Flash ROM

Use the following procedure to write the kernel.

1. Start up WATCHPOINT.
2. Use RESET → GO to start up Redboot.
3. Download the zImage generated in 5.2 to virtual address 0xa0010000.

Resource >> Download

Open the Download dialog box, specify a file as shown below, set the format to binary and the address to R;0xa0010000, and press the Download button.



4. Execute the following commands using Redboot.

```
RedBoot> fis write -f 0x13240000 -l 0x00200000 -b 0xa0010000
```

* Set zImage so that the samba and other files can be shared between the PC for cross development and the PC for image transmission.

Chapter 6 File System Image Construction Method

This section explains the method for constructing a root file system image (jffs2 format) for SandgateVP.

6.1 jffs2 Image Creation

Create the jffs2 format from the root file system tree.
Use root privileges to execute the following commands.

```
# cd /opt/target/rootfs-xscale-060924-2/usr/sbin
# ./mkfs.jffs2 -e 0x40000 --pad=0x03200000 -s 4096 -n -l -r /opt/target/rootfs-xscale-060924-2
  -o /opt/target/kompressor.jffs2
```

The jffs2 image and kompressor.jffs2 are created under /opt/target.
ftp://sources.redhat.com/pub/jffs2/mkfs.jffs2

Download mkfs.jffs2 from the above URL.

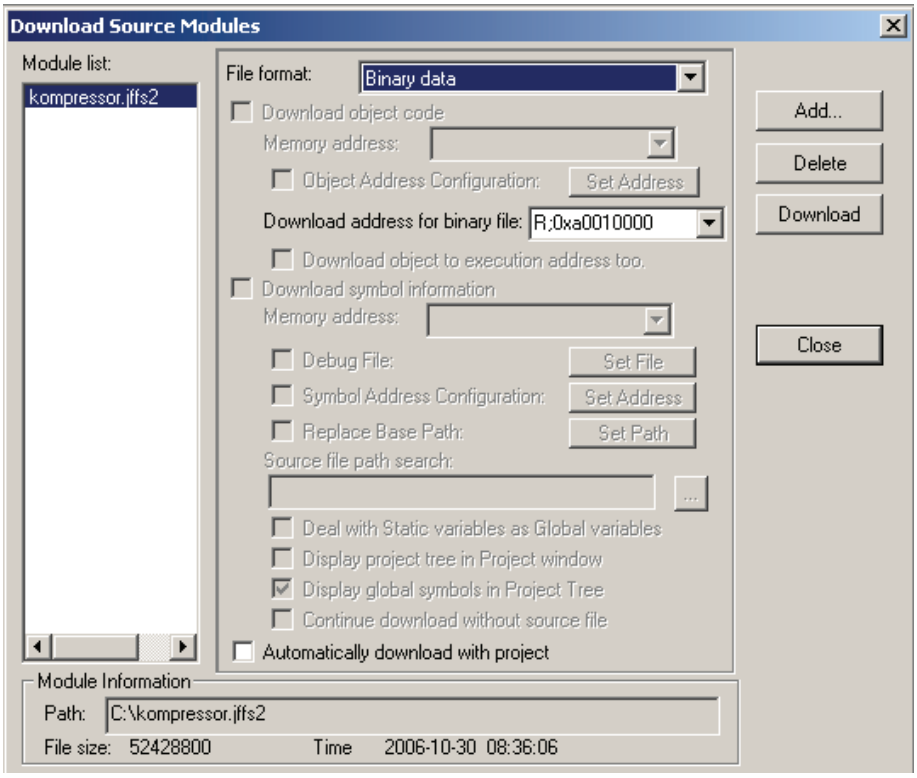
6.2 Method for Writing to Flash ROM

Use the following procedure to write the jffs2 image.

1. Use the WATCHPOINT project file for SandgateVP (SGVP.dbg) to start up WATCHPOINT.
2. Use RESET → GO to start up Redboot.
3. Download the kompressor.jffs2 generated in 6.1 to virtual address 0xa0010000.

Resource >> Download

Open the Download dialog box, specify a file as shown below, set the format to binary and the address to R;0xa0010000, and press the Download button.



4. Execute the following commands using Redboot.

```
RedBoot> fis write -f 0x10040000 -l 0x03200000 -b 0xa0010000
```

* Set kompressor.jffs2 so that the samba and other files can be shared between the PC for cross development and the PC for image transmission.

Chapter 7 Linux Operation Method

This section explains the procedure for starting up Linux using SandgateVP.

7.1 Environment Preparation

Check the following regarding the environment.

PC for console output

- The serial cable is connected.
- The terminal software is running. (Baud rate: 115,200, data: 8-bit, parity: none, stop bit: 1 bit)

SandgateVP

- The serial cable is connected to the serial connection unit for SandgateVP.

7.2 Redboot Startup

If there is no problem with the above environment, turn on the power to SandgateVP.
The following message will be displayed in the terminal software after the power is turned on.

```
+
RedBoot(tm) bootstrap and debug environment [ROM]
Non-certified release, version Tiger.v004-004 - built 13:06:52, Sep 27 2006

Platform: PXA270 Sandgate-VP (XScale)
Copyright (C) 2000, 2001, 2002, Red Hat, Inc.

RAM: 0xa0000000-0xa4000000, 0xa000e1a8-0xa3f9d000 available
FLASH: 0x10000000 - 0x18000000, 512 blocks of 0x00040000 bytes each.
RedBoot>
```

7.3 Kernel Image Load

After the above message is displayed, load the kernel image.
Execute the following commands.

```
RedBoot> fis load zImage
RedBoot> exec -c "console=ttyS0,115200 mem=64M@0xa0000000 root=/dev/mtdblock1
rootfstype=jffs2"
```

When the “exec” command is executed, kernel startup begins and the following message is output.

```
Using base address 0xa1000000 and length 0x00200000
Uncompressing Linux..... done, booting the kernel.
Linux version 2.6.17 (sgvp@x154154) (gcc version 4.1.0) #119 PREEMPT Wed Sep 27 14:06:38 JST
2006
CPU: XScale-PXA270 [69054117] revision 7 (ARMv5TE)
Machine: Sophia Systems Sandgate-VP Development Platform
Memory policy: ECC disabled, Data cache writeback
Run Mode clock: 208.00MHz (*16)
Turbo Mode clock: 416.00MHz (*2.0, active)
Memory clock: 208.00MHz (/2)
System bus clock: 208.00MHz
CPU0: D VIVT undefined 5 cache
CPU0: I cache: 32768 bytes, associativity 32, 32 byte lines, 32 sets
CPU0: D cache: 32768 bytes, associativity 32, 32 byte lines, 32 sets

- Omitted -

pxa27x_udc: version 23-Sep-2004
input: pxa27x-keypad as /class/input/input0
input: sandgatevp keypad driver!
input: pxa27x-pointer as /class/input/input1
input: sandgatevp pointer driver!
TCP bic registered
socket_detect_change
VFS: Mounted root (jffs2 filesystem).
Freeing init memory: 92K
pccard: PCMCIA card inserted into slot 0
pcmcia: registering new device pcmcia0.0
eth0: NE2000 Compatible: io 0xc4860300, irq 42, hw_addr 00:07:40:02:26:C9
pccard: PCMCIA card inserted into slot 1
INIT: version 2.86 booting
```

7.4 Login

Login is possible after the kernel starts up and the login prompt is displayed.
At the login prompt, enter “root” and press the Enter key.

```
(none) login: root
login[1716]: root login    on `ttyS0'

-bash-3.1#
```

Chapter 8 Explanation of Module Functions

This section explains the functions of the Linux modules started up using SandgateVP.

8.1 Key Lighting

8.1.1 Read Function

This acquires the key lighting status.

```
# cat /proc/sys/sandgate-vp/key_light
on
```

When the key lighting is lit, “on” is displayed, and when it is off, “off” is displayed.

8.1.2 Write Function

This turns the key lighting on/off.

```
# echo on > /proc/sys/sandgate-vp/key_light
# echo off > /proc/sys/sandgate-vp/key_light
```

Turn on the key lighting using “echo on,” and turn off the key lighting using “echo off.”

8.2 Vibrator

8.2.1 Read Function

This acquires the key lighting status.

```
# cat /proc/sys/sandgate-vp/vibrator
off
```

When the vibrator is operating, “on” is displayed, and when it is not operating, “off” is displayed.

8.2.2 Write Function

This turns the vibrator on/off.

```
# echo on > /proc/sys/sandgate-vp/vibrator
# echo off > /proc/sys/sandgate-vp/vibrator
```

Turn on the vibrator using “echo on,” and turn off the vibrator using “echo off.”

8.3 Keypad, Pointer

8.3.1 Test Application df_input Startup

```
# df_input
```

The keypad and pointer operation can be checked on the screen.

8.4 Battery Cover Detection

8.4.1 Function

When the battery cover is opened, the following message is output to the console and then the system changes to the Deep-sleep mode.

```
# The assertion of nBATT_FAULT: goto Deep-Sleep Mode
```

8.5 DPM

8.5.1 Script File Settings

This creates a script for using the I/F of the DPM.

```
# vi test_dpm_sample.sh
#!/bin/sh

# Example DPM setup script for Intel XScale PXA27x
#
# 2005 (c) MontaVista Software, Inc. This file is licensed under
# the terms of the GNU General Public License version 2. This program
# is licensed "as is" without any warranty of any kind, whether express
# or implied.
if [ ! -f /sys/dpm/control ]
then
    echo "$0: DPM sysfs filesystem not present."
    echo "May need to mount -t sysfs sysfs /sys"
    echo "or may need to enable CONFIG_DPM."
    exit
fi
echo terminate > /sys/dpm/control
echo init          > /sys/dpm/control
##
## set 'test_dpm' task
##
echo create task_dpm > /sys/dpm/op/control
echo -n 1300       > /sys/dpm/op/task_dpm/v
echo -n 16         > /sys/dpm/op/task_dpm/l
echo -n 3          > /sys/dpm/op/task_dpm/n
echo -n 1          > /sys/dpm/op/task_dpm/b
echo -n 0          > /sys/dpm/op/task_dpm/ht
echo -n 1          > /sys/dpm/op/task_dpm/a
### setting values
## clock
# run/turbo/sysbus/mem=104
#    : l=8, n= 2, a=1, b=1 ht = 0
# run/turbo/sysbus=208, mem=208
#    : l=8, n= 2, a=1, b=1 ht = 0
# run=208, turbo=312, sysbus=208, mem=208
#    : l=16, n= 3, a=1, b=1 ht = 0
# run=208, turbo=416, sysbus=208, mem=208
#    : l=16, n= 4, a=1, b=1 ht = 0
## voltage : v= 1100, 1200, 1300 ,1400 (mv)
```

```
##
## set 'dpm_test' policy
##
echo create dpm_test            > /sys/dpm/policy/control
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task+1
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task+2
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task+3
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task+4
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task-1
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task-2
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task-3
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/task-4
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/idle
echo -n task_dpm                    > /sys/dpm/policy/dpm_test/idle-task
echo -n sleep                   > /sys/dpm/policy/dpm_test/sleep
##
## active 'dpm_test' policy
##
echo -n set dpm_test > /sys/dpm/policy/control
```

Execute the created script.

```
# sh test_dpm_sample.sh
```

Script notation is used to set the voltage and clock.
The above script makes the task_dpm settings to be v = 1300, l = 16, n = 3, a = 1, b = 1, and ht = 0, and the voltage = 13 v, run/turbo/sysbus clock = 208 MHz, and turbo mode clock = 312 MHz.
In addition, the sleep settings are v = 1400, l = 0, n = 0, and sleep = 2, and the CPU switches to the Standby mode.

```
# echo sleep > /sys/dpm/state/active
Stopping tasks: =====|
```

This restores the keypad (except HookSw) and pointer operation from the Standby mode.

```
Restarting tasks... done
```

8.6 Sound

This section explains the method for using a sound device driver to control the sound CODEC and AK4650 incorporated in SandgateVP.

8.6.1 Application Program Creation Precautions

The sound device driver for SandgateVP provides a platform intrinsic ioctl command in addition to the OSS (Open Sound System) standard ioctl command.

A soundcard-vp.h header file is required when creating a sound application. The soundcard-vp.h header file is located under the <directory in which the kernel source is deployed>/include/linux directory. Copy the soundcard-vp.h header file to the usr/include/linux directory of the cross compile environment before compiling the sound application.

Write in the [Format] field of “8.6.3 ioctl System Call” whether or not the soundcard-vp.h of each ioctl command is required.

An audio output program has been provided in CD:¥check_driver¥sound.
For details, refer to CD:¥sample¥sound ¥Readme.txt.

8.6.2 Device File Open

Play and record control is achieved using the read/write/ioctl system call for the /dev/dep file and the ioctl/poll system call for the /dev/mixer file.

Issuing a system call for the /dev/dsp and /dev/mixer file requires that first the file be opened using an open system call.
Writing examples are given below.

- For recording
  fd = open("/dev/dsp", O_WRONLY);
- For playing
  fd = open("/dev/dsp", O_RDONLY);
- For controlling a sound device
  fd = open("/dev/mixer", O_RDONLY);

8.6.3 ioctl System Call

To control a sound device, use the following method to issue an ioctl system call for the /dev/mixer file.

8.6.3.1 Speaker and Receiver Output Control

[Format]

#include <linux/soundcard-sgvp.h>
ioctl(int fd, SANDGATE_VP_SOUND_MIXER_WRITE_ODEVSEL, unsigned int *arg*)

[Explanation]

This is used to control the speaker and receiver output.

To enable the speaker output, specify the SANDGATE_VP_SOUND_ODEV_SPK flag in the argument *arg*. To disable the speaker output, remove the SANDGATE_VP_SOUND_ODEV_SPK flag.
To enable the receiver output, specify the SANDGATE_VP_SOUND_ODEV_RCV flag in the argument *arg*. To disable the receiver output, remove the SANDGATE_VP_SOUND_ODEV_RCV flag.

8.6.3.2 Speaker and Receiver Output Status Acquisition

[Format]

#include <linux/soundcard-sgvp.h>
ioctl(int fd, SANDGATE_VP_SOUND_MIXER_READ_ODEVSEL, unsigned int **arg*)

[Explanation]

This is used to acquire the speaker and receiver output status. The status is stored in the argument *arg*.
When the speaker output is enabled, the SANDGATE_VP_SOUND_ODEV_SPK flag is set in the argument *arg*.
When the receiver output is enabled, the SANDGATGE_VP_SOUND_ODEV_RCV flag is set in the argument *arg*.

8.6.3.3 Speaker and Receiver Mute

[Format]

ioctl(int *fd*, SOUND_MIXER_WRITE_VOLUME, int **arg*)

[Explanation]

To mute the speaker or receiver, 0 is set in the argument *arg*. When muted, the power to the speaker amp is turned off.

To cancel the mute, 100 is set in the argument *arg*. When the mute is cancelled, the power to the speaker amp is turned on.

8.6.3.4 Speaker and Receiver Mute Status Acquisition

[Format]

ioctl(int *fd*, SOUND_MIXER_READ_VOLUME, int **arg*)

[Explanation]

When the speaker and receiver are muted, 0 is stored in the argument *arg*. When not muted, a value other than 0 is stored in the argument *arg*.

8.6.3.5 Headphone Jack Connection Status Acquisition

[Format]

#include <linux/soundcard-sgvp.h>

ioctl(int *fd*, SANDGATE_VP_SOUND_MIXER_READ_HDSET, unsigned int **arg*)

[Explanation]

This acquires the headphone jack insertion status. The status is stored in the argument *arg*.

When a device is connected to the headphone jack, SANDGATE_VP_SOUND_HD SET_CONNECT is set in the argument *arg*.

When a device is not connected to the headphone jack, SANDGATE_VP_SOUND _HDSET_UNCONNECT is set in the argument *arg*.

8.6.3.6 Headphone Mute

[Format]

ioctl(int *fd*, SOUND_MIXER_WRITE_ALTPCM, int **arg*)

[Explanation]

To mute the headphone, 0 is set in the argument *arg*. When muted, the power to the headphone amp is turned off.

To cancel the mute, 100 is set in the argument *arg*. When the mute is cancelled, the power to the headphone amp is turned on.

8.6.3.7 Headphone Mute Status Acquisition

[Format]

ioctl(int *fd*, SOUND_MIXER_READ_ALTPCM, int **arg*)

[Explanation]

When the headphones are muted, 0 is stored in the argument *arg*.

When the headphones are not muted, a value other than 0 is stored in the argument *arg*.

8.6.3.8 Headphone Output Mode Setting

[Format]

ioctl(int *fd*, SNDCTL_DSP_STEREO, int **arg*)

[Explanation]

This sets the headphone output mode (monaural, stereo).

For the monaural mode, 0 is set in the argument *arg*.

For the stereo mode, 1 is set in the argument *arg*.

Set *fd* as the file extension when a /dev/dsp file is opened.

Switch to the monaural mode when the data input from a microphone is played on headphones.

8.6.3.9 Volume Settings

[Format]

ioctl(int *fd*, SOUND_MIXER_WRITE_PCM, int **arg*)

[Explanation]

The Lch (left channel) volume is set in the last 8 bits of the argument *arg*, and the Rch (right channel) volume is set in the first 8 bits of the argument.
The volume is a percentage value between 0 and 100.

bit31	bit24	bit16	bit 8	bit0
		Rch volume	Lch volume	

8.6.3.10 Volume Acquisition

[Format]

ioctl(int *fd*, SOUND_MIXER_READ_PCM, int **arg*)

[Explanation]

The Lch (left channel) volume is stored in the last 8 bits of the argument *arg*, and the Rch (right channel) volume is stored in the first 8 bits of the argument.
The volume is a percentage value between 0 and 100.

bit31	bit24	bit16	bit 8	bit0
		Rch volume	Lch volume	

8.6.3.11 Internal/External Microphone Selection

[Format]

#include <linux/soundcard-sgvp.h>
ioctl(int *fd*, SANDGATE_VP_SOUND_MIXER_WRITE_MICSEL, unsigned int *arg*)

[Explanation]

This is used to select an internal or external microphone.
To select an internal microphone, specify SANDGATE_VP_SOUND_MIC_INT in the argument *arg*.
To select an external microphone, specify SANDGATE_VP_SOUND_MIC_EXT in the argument *arg*.

8.6.3.12 Internal/External Microphone Selection Information Acquisition

[Format]

#include <linux/soundcard-sgvp.h>

ioctl(int *fd*, SANDGATE_VP_SOUND_MIXER_READ_MICSEL, unsigned int **arg*)

[Explanation]

This acquires the microphone switch status. The status is stored in the argument *arg*.

When an internal microphone is selected, SANDGATE_VP_SOUND_MIC_INT is set in the argument *arg*.

When an external microphone is selected, SANDGATE_VP_SOUND_MIC_EXT is set in the argument *arg*.

8.6.3.13 Microphone Mute

[Format]

ioctl(int *fd*, SOUND_MIXER_WRITE_MIC, int **arg*)

[Explanation]

To mute the microphone, specify 0 in the argument *arg*. When muted, the power to the microphone amp is turned off.

To cancel the mute, specify 100 in the argument *arg*. When the mute is cancelled, the power to the speaker amp is turned on.

8.6.3.14 Microphone Mute Status Acquisition

[Format]

ioctl(int *fd*, SOUND_MIXER_READ_MIC, int **arg*)

[Explanation]

When the microphone is muted, 0 is stored in the argument *arg*.

When the microphone is not muted, a value other than 0 is stored in the argument *arg*.

8.6.4 Headphone Jack Insertion Status Change Notification

Notice of the headphone jack insertion status change is given using the poll system call. The poll system call is issued to the /dev/mixer file.

When the headphone jack insertion status changes, the poll system call returns a POLLIN event. After the POLLIN event is generated, issue an ioctl system call (SANDGATE_VP_SOUND_MIXER_READ_HDSET command) to check the headphone jack connection status.

The poll system call usage method is the same as that for normal event operation.

8.6.5 PCM Data Play

To play PCM data, issue a write system call to the /dev/dsp file. The write system call usage method is the same as that of normal file I/O operation.

8.6.6 PCM Data Record

To record PCM data, issue a read system call to the /dev/dsp file. The read system call usage method is the same as that of normal file I/O operation.

Switch the headphone output mode to the monaural mode when the data input from a microphone is played on headphones. For information regarding how to make the switch, refer to “8.6.3.8 Headphone Output Mode Setting.”

8.7 IrDA

This section explains the method for using a device driver to control the IrDA module incorporated in SandgateVP.

8.7.1 IrDA Interface Check

The ifconfig command can be used to check the current network setting status. Use the "-a" option to check all installed interfaces.

An example ifconfig command execution is given below.

```
# /sbin/ifconfig -a
irda0     Link encap:IrLAP  HWaddr 8a:bd:b3:d8
          UP RUNNING NOARP  MTU:2048  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:8
          RX bytes:0 (0.0 b)  TX bytes:0 (0.0 b)

lo        Link encap:Local Loopback
          LOOPBACK  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 b)  TX bytes:0 (0.0 b)
```

8.7.2 IrDA Interface Startup

The ifconfig command can be used to freely start up the IrDA interface. When the IrDA interface has been started up, the IrDA module's shutdown mode is cancelled.
An example IrDA interface startup is given below.

```
# /sbin/ifconfig irda0 up
```

8.7.3 IrDA Interface Shutdown

The ifconfig command can be used to freely shut down the IrDA interface. When the IrDA interface has been shut down, the IrDA module's shutdown mode is entered.
An example IrDA interface shutdown is given below.

```
# /sbin/ifconfig irda0 down
```

8.7.4 Communication Check Procedure

This section explains the communication check procedure for when using 2 SandgateVPs.

Have ready 2 SandgateVPs. For convenience, the respective units are called SandgateVP (A) and SandgateVP (B). SandgateVP (A) is the reception side and SandgateVP (B) is the transmission side.

(1) The IrDA interface runs on both SandgateVP units.
(2) Execute /usr/sbin/irattach irda0 –s 1 on SandgateVP (A).
(3) Execute cat /proc/net/irda/discovery on SandgateVP (A).
(4) Execute /usr/sbin/irattach/irda0 on SandgateVP (B).
(5) Execute the /usr/sbin/irprintf command on SandgateVP (A).
(6) Execute the /usr/sbin/irscanf command on SandgateVP (B).
(7) If the characters input from SandgateVP (B) are displayed on SandgateVP (A), the communication was successful.

The serial console display is shown below.

SandgateVP(A)
-bash-3.1# /sbin/ifconfig irda0 up
-bash-3.1# /usr/sbin/irattach irda0 -s 1
-bash-3.1# cat /proc/net/irda/discovery
IrLMP: Discovery log:

SandgateVP(B)
-bash-3.1# /sbin/ifconfig irda0 up
-bash-3.1# /usr/sbin/irattach irda0
-bash-3.1# cat /proc/net/irda/discovery
IrLMP: Discovery log:

nickname: Linux, hint: 0x0400, saddr: 0xc5c52b19, daddr: 0x4a730e7a

SandgateVP(A)
-bash-3.1# /usr/sbin/irprintf
IrDA printf server starting ...
Waiting for connection!

SandgateVP(B)
-bash-3.1# /usr/sbin/irscanf
Connected!
Type your text with '.' on the last line...

If the test is entered into the Sandgate (B) serial console and the Enter key pressed under the above conditions, the test will be output to Sandgate (A).

If the communication destination unit cannot be detected, adjust the facing distance and angle between the communication ports.

[Execution command]
$ cat /proc/net/irda/discovery

[Results when the communication destination unit could not be detected]
$ cat /proc/net/irda/discovery
IrLMP: Discovery log:

[Results when the communication destination unit could be detected]
$ cat /proc/net/irda/discovery
IrLMP: Discovery log:
nickname: Linux, hint: 0x0400, saddr: 0xc5c52b19, daddr: 0x4a730e7a

8.8 USB

This section explains the method for using the file storage cassette function of the USB port built into the SandgateVP.

8.8.1 File Storage Cassette Function

The file storage cassette function can be used to access a memory card connected to the CF slot as a “removable disk” from a Windows PC, etc.

1. Format the CF memory card in the VFAT format.
2. Install the CF memory card in the SandgateVP.
3. Install the file storage cassette module.

```
# insmod /lib/modules/2.6.17/kernel/drivers/usb/gadget/g_file_storage.ko file=/dev/hda1
stall=0 removable=1
g_file_storage gadget: File-backed Storage Gadget, version: 28 November 2005
g_file_storage gadget: Number of LUNs=1
g_file_storage gadget-lun0: ro=0, file: /dev/hda1
```

4. Use a USB cable to connect the USB ports of the SandgateVP and the Windows PC.
5. The Windows PC recognizes the SandgateVP’s CF memory card as a “removable disk.”

8.9 WLAN

The SGVP’s onboard module is connected to the access point and a check is conducted until a ping is ping is successfully received.

For details regarding this procedure, refer to the CD:¥check_driver¥WiFi¥WiFi_Operation_Instructions.pdf.

8.10 Compact Flash

This checks up to the CF boot and CF mount.

For details regarding this procedure, refer to the CD:¥check_driver¥CF¥CF_Operation _Procedure.pdf.